AEGIS & AEGIS



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Johnson & Johnson Case Competition

MEET THE TEAM



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Accounting; Freshman



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Finance; Freshman



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Finance; Freshman



JAYSUN YOON

Finance; Freshman

Context & The Challenge

Understanding A&A's position before making decisions

A&A 2026

MARKET CONTEXT & THE CHALLENGE

A&A · Wound Closure

MARKET SHIFT

7.5%

Annual Market Growth

- Non-barbed → Barbed sutures
- Robotic surgery removes old barriers
- Knotless design = faster, safer ORs

A&A's POSITION

#1

Non-Barbed Market Position

- Leader in non-barbed sutures
- Cornelia, GA manufacturing plant
- 0 barbed suture products

THE GAP

2

Technologies to Launch

- Symmetrical: IP available to buy
- Spiral: no competitor on market yet
- 5 decisions must align as a system

SWOT Analysis

Understanding A&A's position before making decisions

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STRENGTHS & WEAKNESSES

Internal Factors

S STRENGTHS

Internal · Helpful

Market Leadership

Current market leader with 250+ operating companies

Manufacturing Base

Rooted Infrastructure in Cornelia, GA

Surgeon Trust & Reliability

Decades of clinical trust

W WEAKNESSES

Internal · Harmful

No Barbed Product

Zero presence

No Owned IP

Every entry route costly

Revenue Erosion

Risks eroding non-barbed revenue

OPPORTUNITIES & THREATS

External Factors

O OPPORTUNITIES

External · Helpful

Market Shift to Barbed Sutures

Barbed adoption accelerating via robotics

Acquire Athena

Affordable, IP-eliminating acquisition

Global Distribution

Instant global reach

T THREATS

External · Harmful

Established Rivals

Rivals hold 2–7 year headstart

Substitute Technologies

Substitutes compressing suture market ceiling

Pricing Pressure

Late entry crushes already thin margins

Decisions & Rationale

What we chose, why we chose it, what we ruled out

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DECISION OVERVIEW

Five Decisions | One Integrated Strategy

Securing today's market position, while building tomorrow's dominance

	DECISION		WHY WE CHOSE IT
01	**R&D Strategy**	**Internal (Spiral) + IP Purchase (Symmetrical)**	✓ Builds proprietary moat on spiral; buys speed-to-market on symmetrical.
02	**Pricing**	**Medium**	✓ Keeps 80% of hospitals within reach while maintaining the margins
03	**Marketing**	**Gold**	✓ Builds the surgeon trust that drives inclusion and purchasing decisions.
04	**Supply Chain**	**Outsource**	✓ Outsourcing eliminates CapEx and launch delays
05	**Human Capital**	**Base (Required by Outsource SC)**	✓ Base HC is both required by the case rules under outsourcing

R&D Strategy

IP or Internal Development

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R&D STRATEGY

Decision 01

SPIRAL → INTERNAL DEVELOPMENT

First to Market

No spiral suture exists on the market

- Owned IP — no royalty obligations
- Earlier to market → higher share
- R&D costs amortize over product life

SYMMETRICAL → IP ACQUISITION

Rapid-Entry to Market

Athena's Kronos IP - FDA-cleared since 2005

- Immediate market entry vs. rivals
- One-time cost vs. multi-year R&D burn
- Frees R&D bandwidth for spiral work

✗ Full internal for both = symmetrical delayed 2+ yrs, losing share to Crius/Athena · Full IP for both = royalty drag on A&A's highest-potential product (spiral)

Introducing

FDA CLEARED

BarboFlex™ Suture System

Knotless wound closure for every surgical need

AVAILABLE NOW

BarboFlex™ S

Symmetrical Barbed Suture

"Reliable knotless closure - built for every OR, every surgeon."

- Unidirectional — simple, intuitive technique for any trained surgeon
- Knotless design reduces OR closure time and post-op infection risk



IDEAL FOR

General surgery | Orthopedics | Cardiovascular | Wound closure

FIRST TO MARKET · REGISTER INTEREST

BarboFlex™ X

Spiral Barbed Suture System

"The future of wound closure — engineered for the robotic OR."

- Shorter, less invasive procedures — reduced patient recovery time
- Developed and owned by A&A — the only spiral suture of its kind



IDEAL FOR

Robotic surgery | Laparoscopic procedures | Complex wound closure

Pricing Strategy

Low / Medium / High

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PRICING STRATEGY

Decision 02 · Medium Selected

LOW

100%

Target market

Below market avg price

▲ Max share

▼ Erodes margin; hard to reprice

Ruled out

MEDIUM

80%

Target market

Market average price

▲ Share + margin balance

▼ Requires strong marketing support

Selected ✓

HIGH

65%

Target market

Above market avg price

▲ Max margin

▼ Low share; risks competitor entry

Ruled out

Medium → Market share: 85.8% | Gross margin: 76%

Marketing Strategy

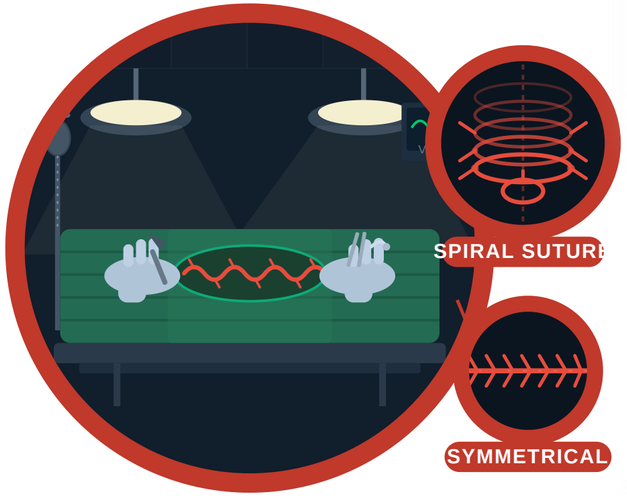
Bronze / Silver / Gold

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MARKETING STRATEGY

AEGIS & AEGIS · MEDTECH SURGERY

Advanced
Wound Closure Portfolio



80%
of key platforms
#1 or #2 position

KNOTLESS DESIGN
barbed · water-tight

ROBOTIC
surgery ready

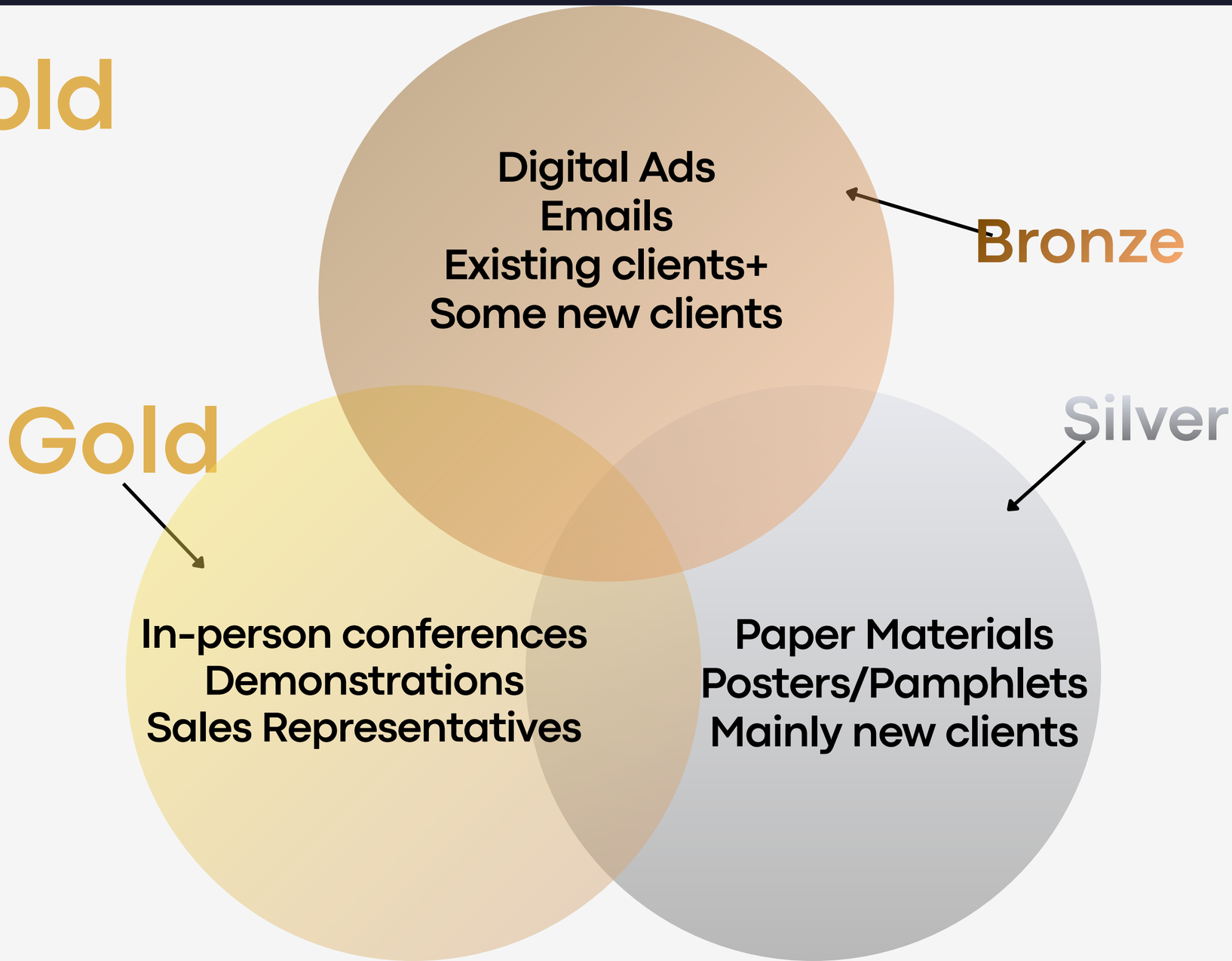
The Future of Surgery

Trust
- Surgeon confidence
- Clinical Validation
- Proof of success

Market
- Targeting professionals
- Small market
- Fine-tuned

Mixed Strategy
- Short term
- Long term
- Education

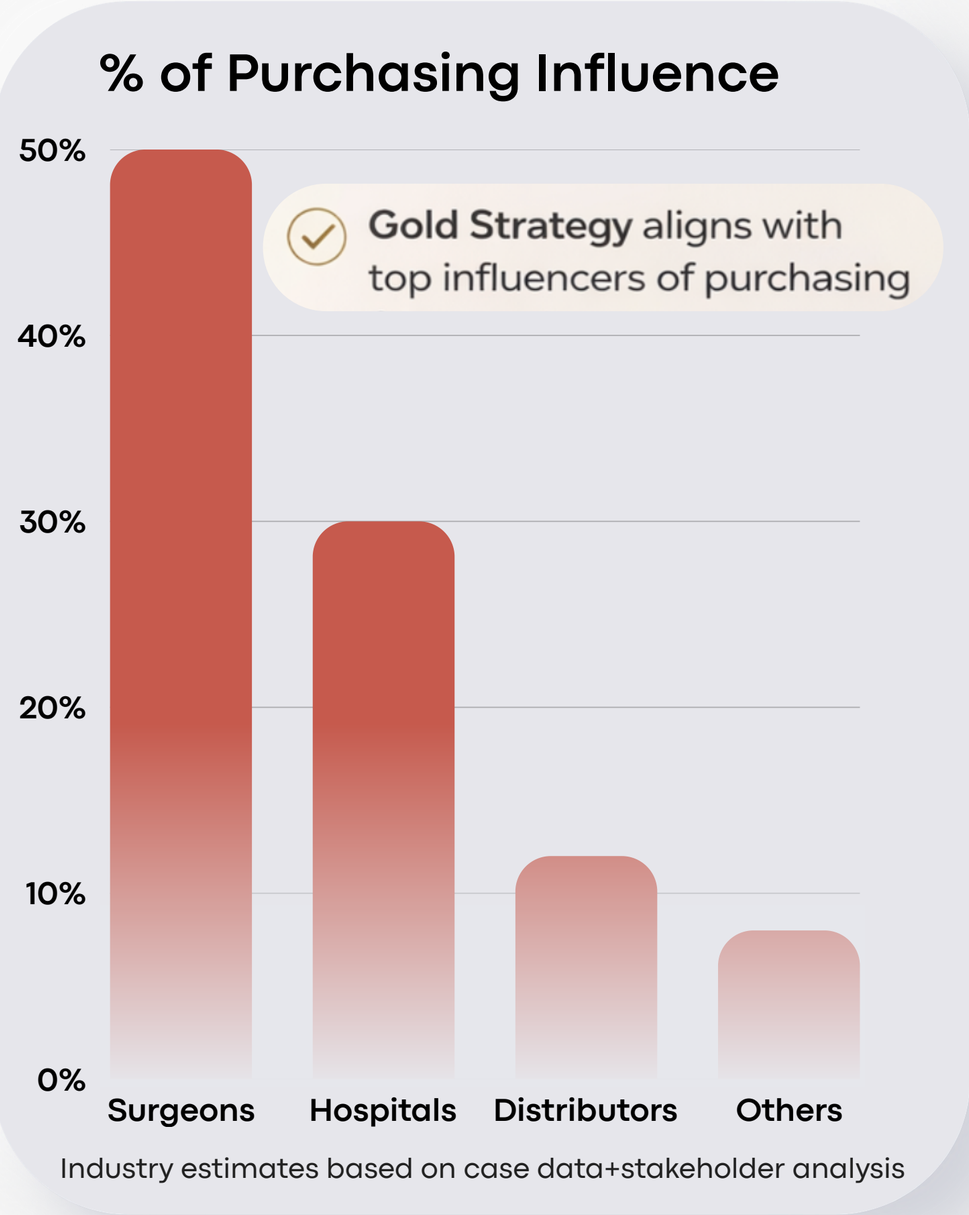
Why Gold Wins



Targeted Market

- Need surgeon familiarity + trust
- Relationship-driven, not marketing-driven

80% of surgical platforms where A&A holds top #1-2 positions



Supply Chain & Human Capital

Outsource / Plant Expansion / New Plant

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SUPPLY CHAIN & HUMAN CAPITAL

Decisions 04 & 05

Outsource ✓

Partner with external manufacturer

- Lowest CapEx
- Immediate revenue
- Variable COGS ~18.9%

Plant Expansion

Expand Cornelia, GA facility

· Moderate CapEx
· 3-year delay
· Revenue from Yr 4

New Plant

Build dedicated barbed facility

· Highest CapEx
· 5-year delay
· Revenue from Yr 6

HUMAN CAPITAL · BASE (Required — Outsource SC eliminates the HC decision)

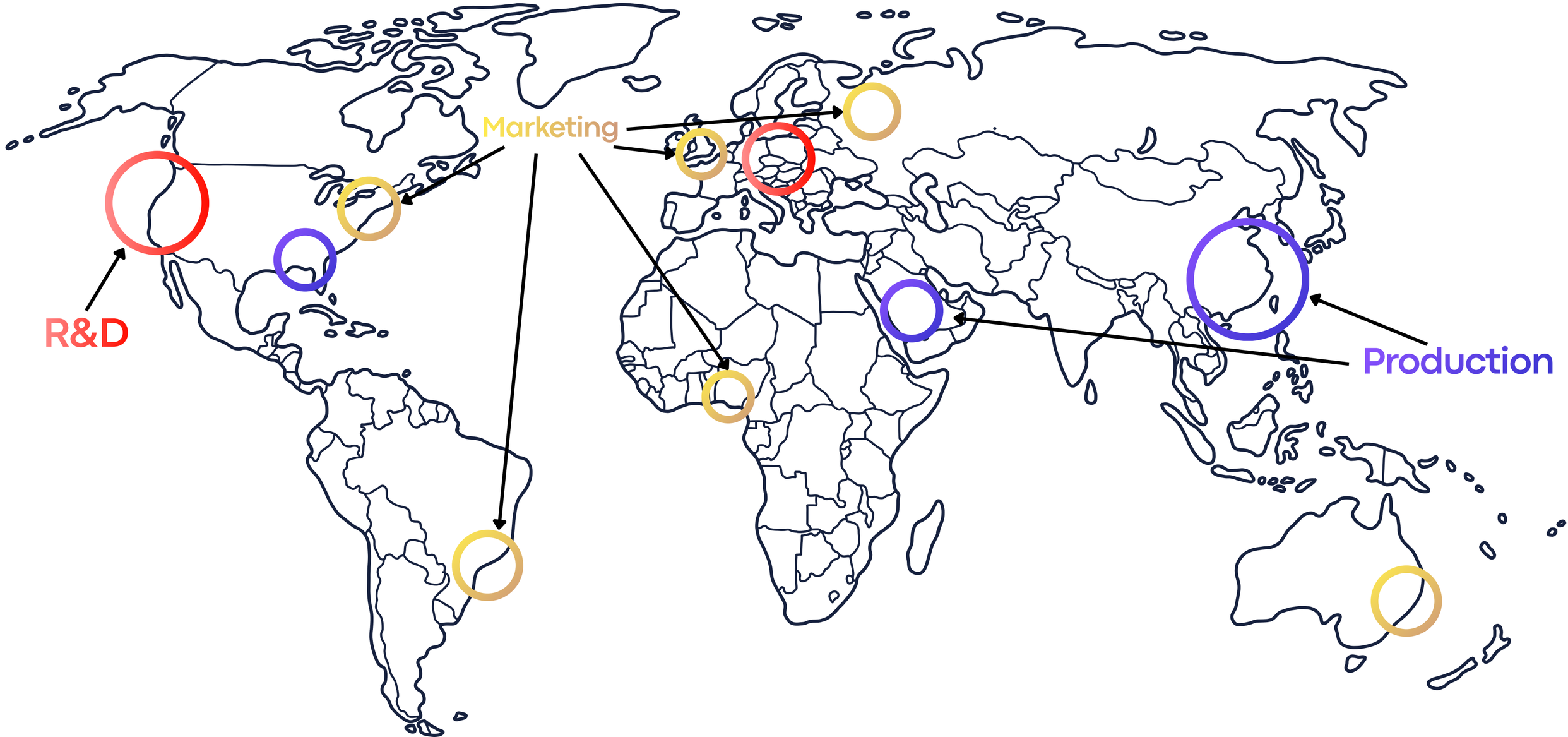
33%→20%
SG&A % NTS Yr 0→Yr 10

No CapEx
HC hiring deferred

Scalable
Floor, not a ceiling

Outsource frees capital for IP acquisition and Gold marketing — the two highest-ROI decisions in the model.

A&A's Global Infrastructure



Financial Analysis

How decisions translated into value

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KEY FINANCIAL METRICS

11%

A&A Growth Rate

Outpacing Market

56.4%

Beginning Market Share

Start position Year 0

85.8%

Peak & ending Share

+29.4pt gain over 10 years

4 years

Payback period

CapEx=discounted cash flow

$116B

Cumulative net sales

High sales*medium pricing combination

76%

Cumulative gross margin

$88.2B gross profit

$23.2B

NPV@ 7%

+10.6B baseline

IRR > 7%

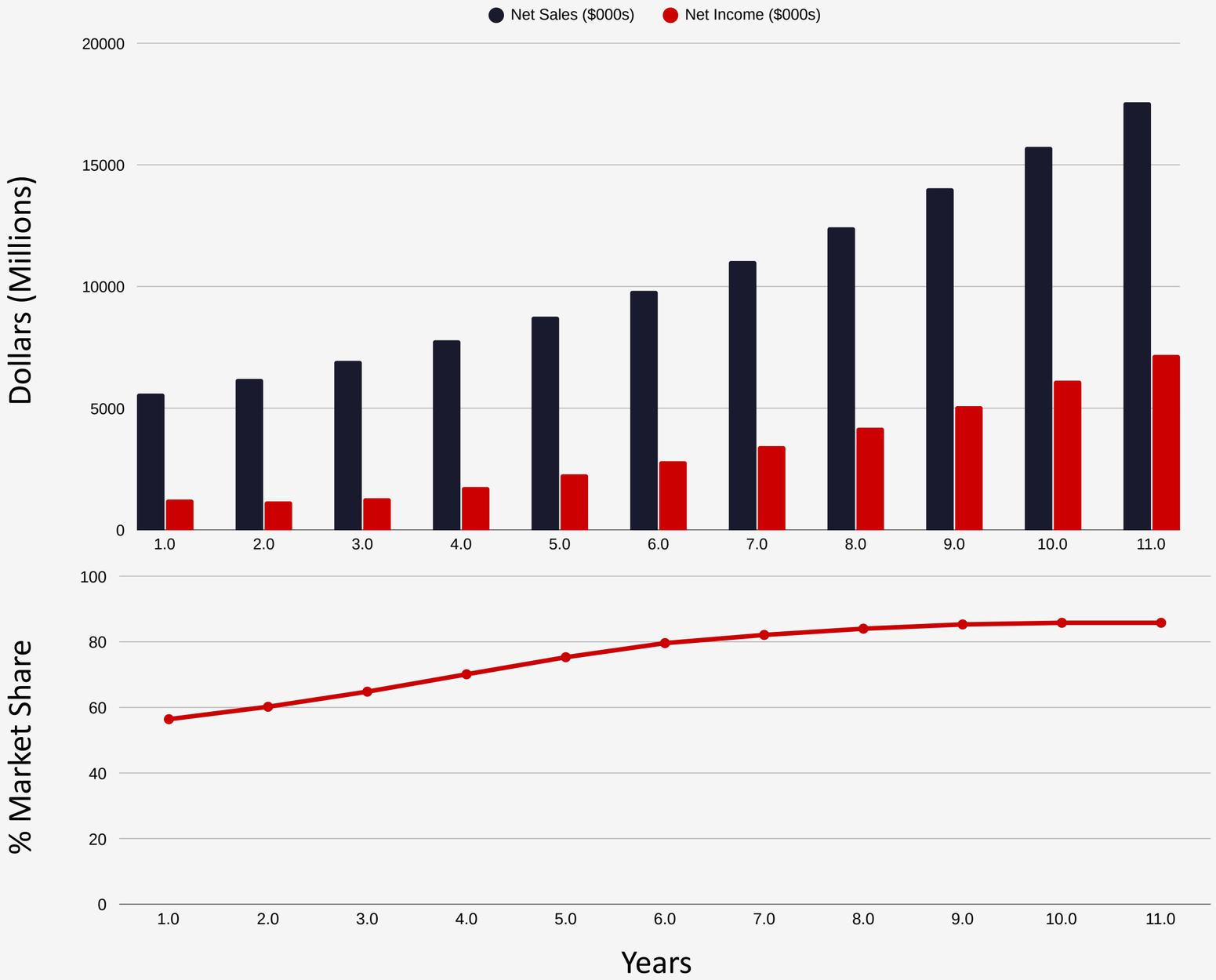
Clears cost of capital

Strategy creates value for stockholders

Assumptions: 7.5% market growth · 7% WACC · 24% tax rate · Beginning share 56.4% · Financials are model outputs, not actuals

REVENUE, NET INCOME & MARKET SHARE

Financial Analysis 2 of 3



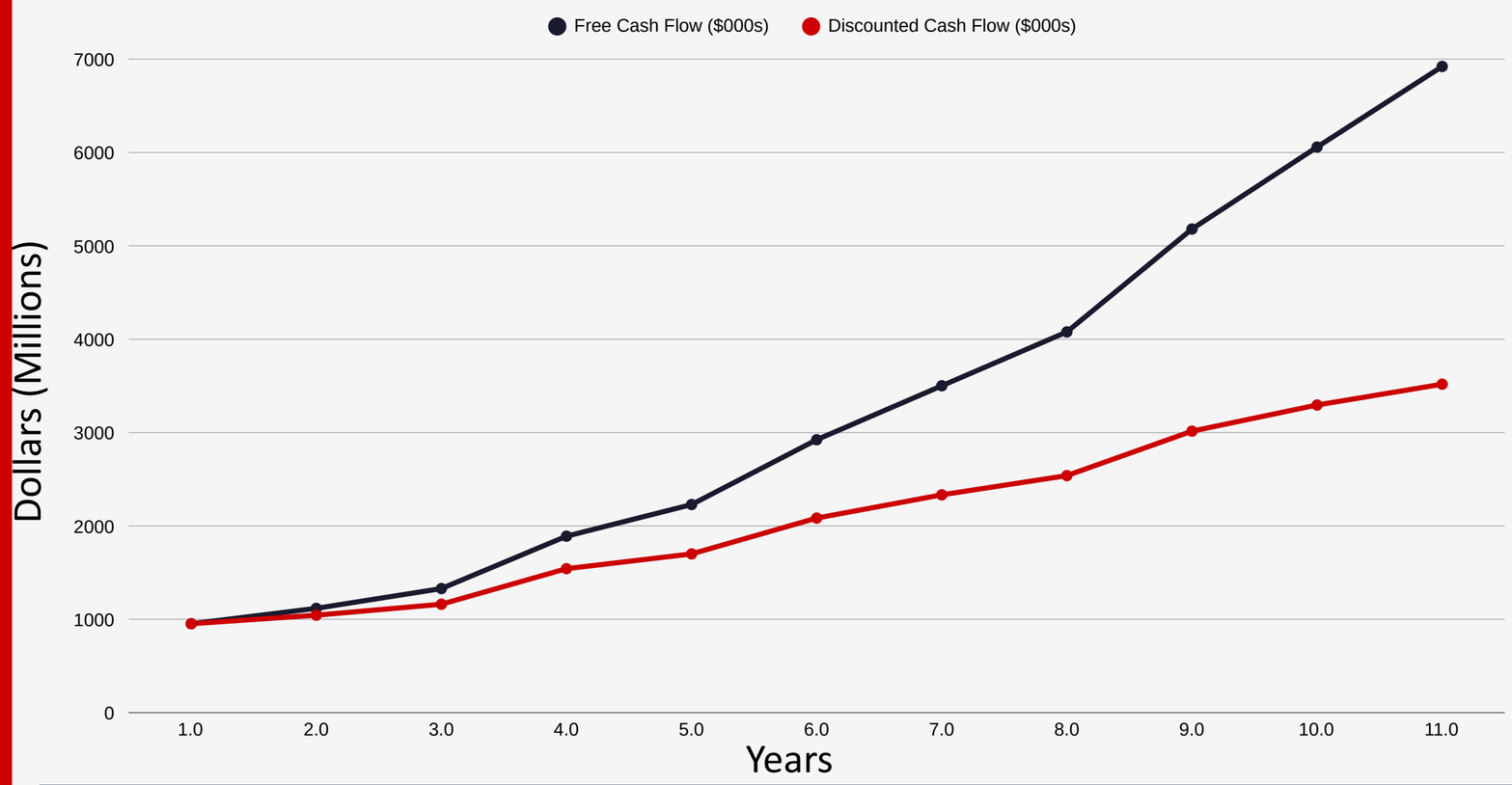
~11%
Revenue CAGR
$5.6B → $17.6B

+29%
Share Gain
56.4% → 85.8%

69%→81%
Gross Margin
12% expansion

22%→41%
Net Margin
Operating leverage

CASH FLOW & DECISION LINKAGE



$23.2B

NPV @ 7%

FCF: $952M → $6.9B

Cumulative FCF: $36.2B

Discount Rate: 7%

DECISION → FINANCIAL IMPACT

Internal Spiral	▸ +$2.4B NPV | Owned IP → no royalty drag on highest-volume product → supports 76% avg gross margin
IP Symmetrical	▸ +0.88B NPV | Immediate market entry → beginning share 56.4% vs. lower with internal development
Medium Pricing	▸ +3.92B NPV| 80% market reach at healthy ASP → GP 69% (Y0) climbing to 81% (Y10)
Gold Marketing	▸ $1.6B NPV | 29pt share gain → $116B cumulative revenue vs. significantly less with Silver/Bronze
Outsource SC	▸ +$2.4B NPV | COGS variable ~18.9%

Sensitivity Analysis

Analyzing the competition

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SENSITIVITY ANALYSIS

NPV Impact of Decision Changes

NPV remains positive and relatively stable under all primary downside scenarios

VARIABLE	BASE	DOWNSIDE	UPSIDE	BASE NPV	DOWN NPV	UP NPV	KEY INSIGHT
Marketing Tier	**Gold**	Silver	N/A	**$23.2B**	~$21.8B	N/A	$1.4B NPV at risk - confirms Gold is non-negotiable
Pricing	**Medium**	Low	High	**$23.2B**	~$19.8B	~$24.7B	Medium is NPV optimum, accounting for A&A values
R&D: Symmetrical	**IP Purchase**	Internal	N/A	**$23.2B**	~$20.8B	N/A	Internal dev delays entry — rivals deepen hospital relationships during the gap
R&D: Spiral	**Internal**	IP Purchase	N/A	**$23.2B**	~$21.4B	N/A	IP purchase adds royalty drag. Internal is superior long-term
Supply Chain	**Outsource**	New Plant	N/A	**$23.2B**	~$22.1B	N/A	Plant expansion CapEx costs ~$3M NPV + 3yr revenue delay

Critical path: Pricing has the single largest NPV downside if changed. Market growth rate is the most sensitive external variable.

Competitive Analysis

How A&A's portfolio competes in the barbed suture market

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COMPETITIVE ANALYSIS

A&A vs. Others

Why A&A wins: Only player launching both technologies simultaneously with relationships Crius and Athena cannot match.

COMPANY	PRODUCTS	PRICE	MKT	STRENGTH	WEAKNESS	STATUS
A&A (Our Strategy)	Spiral (internal, first-to-market) + Symmetrical (Kronos IP)	Medium	Gold	Dual-tech + brand + global scale	Premium cost structure	**LEADER**
Crius (Poseidon)	Poseidon - $11.9B revenue.	–	Est.	Market since 2010, strong hospital base	No spiral product	**RIVAL**
Athena (Kronos)	Kronos - Patent 1999 $221M	–	Targeted	Early IP + bidirectional portfolio	Small scale vs. A&A	**RIVAL**
Others (Hades / Apollo)	Preclude AJA, GAIA	Var	Niche	Technology disruption potential	Limited OR penetration	DISRUPTOR

Values-Driven Strategy

Credo: "We believe our first responsibility is to the patients, doctors and nurses, to mothers and fathers and all others who use our products and services."

WHO WE SERVE	HOW WE SERVE
Patients	Proprietary & Acquired Sutures
Employees	Outsourced Manufacturing
Community	Global Expansion
Stockholders	NPV-Positive Strategy

Why This Strategy Wins

01 **Spiral Dominance**

No rival product exists. Internal dev puts A&A first in the premium segment.

02 **Volume + Margin**

Medium + Gold is the only combination driving 85.8% share AND 32% margins.

03 **Efficiency**

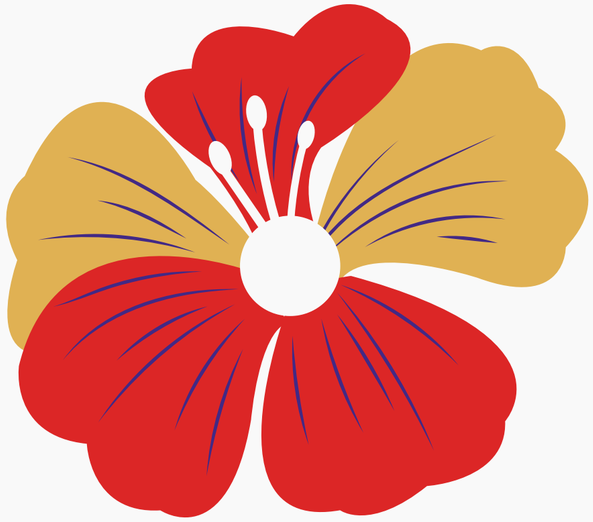
Outsource + Base HC redirects capital to IP and marketing for the highest ROI decisions.

04 **Value**

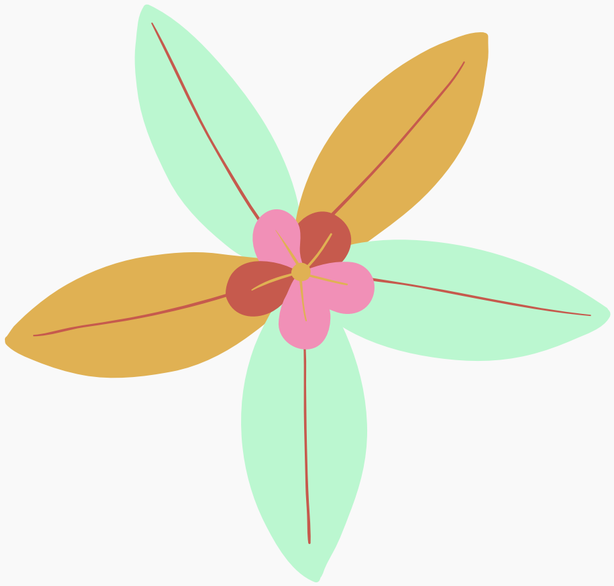
$23.2B NPV. FCF grows 7x. Compounding earnings, not a one-cycle spike.

05 **Scalable**

Outsource SC and Base HC are floors. Model flexes up if market outperforms.







THANK YOU!